



Solution Set

Question 1

$$\lim_{x \rightarrow -2} \frac{\frac{1}{x} + \frac{1}{2}}{x+2} = \lim_{x \rightarrow -2} \frac{x+2}{2x(x+2)} = -\frac{1}{4}$$

Question 2

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

Question 3

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$$

Question 4

$$\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1$$

Question 5

Use algebraic simplification, cancellation, and standard limits before substitution.

Question 6

$$\frac{d}{dx} x^n = nx^{n-1}$$

Question 7

$$\frac{d}{dx} e^x = e^x, \quad \frac{d}{dx} \log x = \frac{1}{x}$$

Question 8

$$\frac{d}{dx} \{f(x)g(x)\} = f'(x)g(x) + f(x)g'(x)$$

Question 9

$$\frac{\frac{d}{dx} f(x)}{g(x)} = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$$

Question 10

$$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$$

Question 11

For a local optimum, solve

$$f'(x) = 0$$

and classify with

$$f''(x) > 0 \Rightarrow \text{minimum}, \quad f''(x) < 0 \Rightarrow \text{maximum}$$